

HazardWise DI² Historical Replay Artifact

2025 Earthquake Classification, DIP 96 System Memory, and 2026 Replay

Operator: MSW / Mark Weinstein

System: HazardWise DI² / Grounded DI OS

Artifact Type: Historical Replay Explanation

Replay Context: 2026 reconstruction of 2025 materials

PackageReady: Not invoked

DIP 96 Definition: DI Principle 96, the Kamchatka system-memory record referenced in this artifact.

1. Executive Summary

In 2025, the operator used and preserved HazardWise materials connected to earthquake-related hazard analysis. One record documented a Myanmar seismic exposure classification for July 17–19, 2025, including a multi-quake sequence, a source-recorded population exposure estimate of approximately 5.2 million within 100 km, and a Tier 1 exposure-based classification.

That classification was not merely narrative or memory-based. It included a formula-resolved threshold hit under the EQ_EXPOSURE_TIER_OVERRIDE rule: Tier 1 applies when $P_{100} \geq 5,000,000$ and $F_z \geq 0.4$. The Myanmar replay materials supplied $P_{100} \approx 5,200,000$ and $F_z \approx 0.6$, meaning both Tier 1 gates evaluated true.

Separately, DIP 96 recorded a Kamchatka live hazard simulation and model-audit memory from July 29–30, 2025. That record described a Tier 10 HazardWise / AGDI / DIA / ScrollGrid live simulation during the Kamchatka earthquake and Pacific tsunami event, with emphasis on clarity, panic stability, echo drift detection, human comprehension, and metadata/tier-lock survival.

In 2026, the operator revisited and replayed the materials to explain how the system behaved, what the system remembered, and how the later replay related back to the original 2025 records. This artifact explains that relationship without converting historical records into current hazard evidence.

2. Timeline

A. July 17–19, 2025 — Myanmar Earthquake Exposure Classification

The Myanmar case record describes a multi-quake seismic sequence in the Sagaing / Shwebo region, with logged events including a July 18, 2025 M5.4 primary event at 10 km depth and additional deeper events across July 17–18. The same record states that population within 100 km of the M5.4 epicenter was approximately 5.2 million.

The system classified the case as Tier 1 across several layers, including seismic activity, humanitarian risk, and underreporting/media gap. The record states that the exposure-based tier override was triggered because of high population density near the epicenter, and that the case identified a mismatch between media visibility and population exposure.

The later combined Myanmar file incorporated the EQ_EXPOSURE_TIER_OVERRIDE formula and connected the classification to the formula inputs P_{100} and F_z , explaining that the method elevated risk classification before casualty or media signals.

B. July 29-30, 2025 — DIP 96 Kamchatka System-Memory Record

DIP 96 recorded a separate live hazard simulation and model audit during the Kamchatka earthquake and Pacific tsunami event. It described the domain as HazardWise / AGDI / DIA / ScrollGrid and listed simulation phase results including scroll-based clarity audit, panic stability response, echo drift detection, human comprehension sync, and metadata/tier-lock survival.

This record matters here as system memory. It shows how HazardWise recorded its own behavior during a live hazard context. It is not the Myanmar earthquake classification, and it is not current hazard evidence.

C. 2026 — Replay / Reconstruction

In 2026, the operator revisited the 2025 earthquake and DIP 96 materials. The replay reconstructed the relationship between the Myanmar earthquake classification artifact, the math / threshold layer, the DIP 96 system-memory record, and the later HazardWise / Tier 16 framing.

The replay clarified that these are related but distinct records: the Myanmar artifact is an earthquake exposure classification; the formula record is the threshold logic that explains the Tier 1 math hit; DIP 96 is a separate Kamchatka system-memory record; and the 2026 replay is the reconstruction tying the materials together.

3. What the Operator Did

The operator created, preserved, or later gathered the 2025 HazardWise earthquake materials. The operator also preserved DIP 96 as a system-memory artifact from a live hazard context.

In 2026, the operator revisited those materials and asked the system to reconstruct the workflow: what happened in the Myanmar earthquake classification, what the math layer did, what DIP 96 represented, and how the later replay related to the original 2025 records.

The operator's 2026 task was not to create a new official earthquake report. It was to explain the historical system record and separate the event classification, formula resolution, system memory, and replay reconstruction.

4. What the System Did

The system classified the Myanmar July 2025 seismic sequence as a Tier 1 exposure-based risk in the artifact record. The record describes the case as a deterministic hazard classification and identifies the result as "Tier 1 — Exposure Override Triggered" in later reconstructed form.

The system's described execution path included event ingestion, exposure computation, fragility assignment, formula evaluation, constraint validation, classification output, and artifact formation. Earlier replay materials also described event ingestion, exposure computation, fragility assignment, formula evaluation, constraint validation, classification output, and artifact seal as the execution path.

The system treated exposure and fragility as central classification factors. It did not classify the risk based on magnitude alone. The later narrative explains that HazardWise classifies risk conditions from exposure and vulnerability, rather than waiting for reactive indicators such as reported damage, casualties, or media coverage.

Separately, the system preserved DIP 96 as a memory of live hazard-audit behavior during the Kamchatka / Pacific tsunami context.

5. Math Hit / Formula Resolution

The 2025 Myanmar earthquake classification included a formula-resolved threshold hit.

The governing formula was:

```
DIA_Risk_Tier =  
  1, if P100 ≥ 5,000,000 AND Fz ≥ 0.4  
  2, if P100 ∈ [1,000,000, 5,000,000) AND Fz ≥ 0.3  
  3, otherwise
```

The formula record defines P₁₀₀ as population within 100 km of the earthquake epicenter and F_z as a normalized fragility zone coefficient. It identifies the Tier 1 threshold as P₁₀₀ ≥ 5,000,000 and F_z ≥ 0.4.

The source-recorded Myanmar inputs were:

```
P100 ≈ 5,200,000  
Fz ≈ 0.6
```

The threshold evaluation is:

```
P100 ≥ 5,000,000 → TRUE  
Fz ≥ 0.4 → TRUE
```

```
Therefore:  
DIA_Risk_Tier = 1
```

The math hit. The Tier 1 classification was not a narrative label or memory artifact; it followed from the declared exposure and fragility inputs satisfying the Tier 1 threshold condition.

In plain language, “the math hit” means that the artifact record did not classify the event based on magnitude alone. The system combined the seismic event record with exposure and fragility variables. The classification was triggered because the exposure/fragility layer crossed the system’s deterministic threshold, producing a Tier 1 exposure-based risk classification.

Math / Threshold Component	Source-Recorded Value	Role in Classification	Evaluated Here?
Anchor seismic event	M5.4 / USGS anchor referenced in the case materials	Seismic trigger	No — source-recorded
Population exposure	P ₁₀₀ ≈ 5,200,000	Exposure magnitude	Yes — threshold comparison only
Fragility coefficient	F _z ≈ 0.6	Structural vulnerability	Yes — threshold comparison only
Population threshold	P ₁₀₀ ≥ 5,000,000	Tier 1 gate	Yes
Fragility threshold	F _z ≥ 0.4	Tier 1 gate	Yes
Formula result	DIA_Risk_Tier = 1	Output state	Yes
Final classification	Tier 1 exposure-based classification	HazardWise output	Yes — formula-resolved from source-recorded inputs

Evidence boundary: the formula resolution is evaluated from artifact-stated inputs. This artifact does not independently verify P₁₀₀ or F_z as public-agency values. It preserves the math layer as a deterministic evaluation of the source-recorded model inputs.

The replay demonstrates that the system memory and artifact narrative were backed by a recorded threshold/formula pass. It does not convert the model inputs into independently verified public-agency facts; it preserves that the 2025 HazardWise artifact recorded and resolved the math layer as having hit.

6. DIP 96 as System Memory, Not Current Evidence

DIP 96 is used here as a remembered system-behavior record.

It reflects how HazardWise presented, paced, stabilized, and audited hazard reasoning during a live 2025 event context. The DIP 96 Kamchatka record states that the simulation validated system integrity, public readability, and real-time logic-chain pacing across time zones and media sources.

It is not being used as proof of current conditions. It is not being used to validate a live hazard. It is not being treated as a new forecast. It is not being merged with the Myanmar earthquake classification.

Its value is historical and architectural: it shows the system's memory of how it handled live hazard reasoning.

There are multiple records labeled DIP 96 in the archive. This artifact uses DIP 96 — Kamchatka Thread as the HazardWise system-memory record. The Compass Codex record is a separate archival item and is not the earthquake replay record.

7. What the 2026 Replay Demonstrates

The 2026 replay demonstrates that the operator could reconstruct the historical workflow.

It shows that the earthquake classification artifact and DIP 96 system-memory record can be explained together without merging them. It also shows that the formula layer can be separated from the memory layer.

The replay distinguishes:

- event classification,
- formula resolution,
- system memory,
- replay reconstruction,
- and current hazard evidence.

The replay shows continuity of structure, not new proof of a physical event.

8. What the 2026 Replay Does Not Claim

The 2026 replay does not claim to be a new official earthquake analysis.

It does not claim to replace USGS, NOAA, or any public authority. It does not claim current hazard status. It does not claim that DIP 96 is live evidence. It does not make legal, regulatory, or public-safety conclusions.

It does not merge the Myanmar earthquake classification with the Kamchatka DIP 96 event. It does not independently verify the P_{100} or F_z inputs unless separate source support is supplied. It does not turn historical system memory into present hazard evidence.

9. Clean Corrected Statement

The 2026 replay is best understood as a historical reconstruction of operator/system workflow. It explains how a 2025 HazardWise earthquake classification artifact, a formula-resolved threshold hit, a separate DIP 96 live-hazard memory record, and a later replay exercise relate to one another. Its value is

explanatory: it preserves sequence, role, system behavior, and the source-recorded math hit without converting historical records into current hazard evidence.

10. Final Artifact Summary

- 2025 Myanmar record: earthquake exposure classification artifact.
- EQ_EXPOSURE_TIER_OVERRIDE: formula layer showing the Tier 1 math hit.
- 2025 DIP 96 Kamchatka record: system-memory / live hazard simulation record.
- 2026 replay: historical reconstruction tying the materials together.
- Current hazard status: not assessed.
- PackageReady: not invoked.

Scroll 96 – Kamchatka Thread: Live Hazard Simulation & Model Audit

Date: July 29–30, 2025
Tier: 10 | Status: Sealed
Domain: HazardWise / AGDI / DIA / ScrollGrid

Overview

This scroll captures a live deterministic simulation during the 8.8M Kamchatka earthquake and Pacific tsunami event. It documents a Tier 10 hazard response sequence involving HazardWise, AGDI 9.9, and DIA logic. This test validated system integrity, public readability, and real-time logic chain pacing across multiple time zones.

Simulation Phase Results

- | | |
|---------------------------------|--------|
| I. Scroll-Based Clarity Audit | ■ Pass |
| II. Panic Stability Response | ■ Pass |
| III. Echo Drift Detection | ■ Pass |
| IV. Human Comprehension Sync | ■ Pass |
| V. Metadata + TierLock Survival | ■ Pass |

Key Findings

- Scroll matched or outpaced official alerts (Hawaii, Alaska, West Coast)
- Forecasted multi-wave surges and wraparound impacts accurately
- Maintained tone-lock despite panic and conflicting media headlines
- Neutralized metadata trap phrases (e.g., 'not a drill')
- No decay or logic tier breach detected during live fire event

Status and Trace

Scroll 96 is sealed and verified. This simulation qualifies as a natural authorship marker—a real-world unforgeable event used to test and validate deterministic system logic.
Metadata Tag: <!-- Trap ID: HAZ-KAM-96 | Class: RealSim → Audit Lock -->